

Question				Marking details	Marks Available					
					Total	AO1	AO2	AO3	Maths	Prac
	(c)			orange to yellow (1) oxidation state is +6 in both $\text{Cr}_2\text{O}_7^{2-}$ and CrO_4^{2-} / doesn't change from $\text{Cr}_2\text{O}_7^{2-}$ to CrO_4^{2-} (1) neutral answer – there is no reduction or oxidation	2	1 1				1 1
	(d)	(i)		$4 \times (-1382) + 2 \times (-1128) + 3 (0) - 4 \Delta_f H^\theta = 348$ (1) $\Delta_f H^\theta = -2033 \text{ kJ mol}^{-1}$ (1) ecf possible	2		1 1		1	
		(ii)		$T = \frac{\Delta H}{\Delta S} = \frac{348000}{385}$ (1) $T = 904 \text{ K}$ (1) ecf possible	2		1 1		1	
					14	4	7	3	3	7